

Nonlinear and implicit port-Hamiltonian models of sloshing in 1D: structure-preserving numerical methods

Ahlam Ouardi¹

Supervisors:

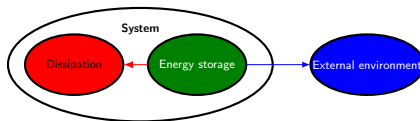
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What are Port-Hamiltonian systems



(i) Energy storage

$$H(x) \geq 0, \quad P_{\text{stored}} = \frac{dH}{dt}$$

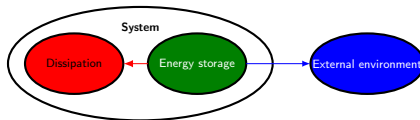
(ii) Dissipation

$$P_{\text{diss}} = \langle e_r, f_r \rangle \geq 0$$

(iii) Environment

$$P_{\text{ext}} = u^\top y$$

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(ii) Dissipation

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$$P_{\text{ext}} = u^\top y$$

PH representation

$$\underbrace{\begin{bmatrix} \dot{x} \\ f_r \\ -y \end{bmatrix}}_f = \underbrace{J}_{\text{Skew-adjoint}} \underbrace{\begin{bmatrix} \delta_x H(x) \\ e_r \\ u \end{bmatrix}}_e$$

(i) storage ports: x energy and $\delta_x H(x)$ coenergy variables
(ii) dissipative ports: $e_r = S f_r$ resistive relation, $S \geq 0$
(iii) external ports: y output, u input

Power balance

$$\text{Skew-adjointness of } J \implies \langle e, f \rangle = 0 \implies \boxed{\frac{dH}{dt} + \langle e_r, f_r \rangle = u^\top y}$$

Why port-Hamiltonian modeling?

- ① **Multi-physics:** mechanics, fluid dynamics
- ② **Energy exchanges:** storage, dissipation and interaction with the environment.

Why structure-preserving discretization?

Preserving this structure allows discretizing subsystems independently and coupling them later while preserving the energy balance.

Objectives

- ① Derive a port-Hamiltonian formulation of the 1D viscous shallow-water equations.
- ② Extend the formulation to the coupled fluid-tank dynamics.
- ③ Construct a structure-preserving finite-dimensional model.
- ④ Perform numerical simulations of viscous sloshing dynamics using SCRIMP ¹.

¹<https://g-haine.github.io/scrump>

- **Viscous shallow-water modeling**
Gerbeau, J.-F. and Perthame, B. (2001). *Derivation of viscous Saint-Venant system for laminar shallow water; numerical validation*. Discrete and Continuous Dynamical Systems – B.
- **Infinite-dimensional port-Hamiltonian systems**
van der Schaft, A.J. and Maschke, B.M. (2002). *Hamiltonian formulation of distributed-parameter systems with boundary energy flow*. Journal of Geometry and Physics.
- **Port-Hamiltonian modeling of sloshing**
Cardoso-Ribeiro, F.L., Matignon, D. and Pommier-Budinger, V. (2017). *A port-Hamiltonian model of liquid sloshing in moving containers and application to a fluid-structure system*. Journal of Fluids and Structures.
- **Structure-preserving discretization – PFEM**
Cardoso-Ribeiro, F.L., Matignon, D. and Lefèvre, L. (2021). *A partitioned finite element method for power-preserving discretization of open systems of conservation laws*. IMA Journal of Mathematical Control and Information.
- **Numerical simulation – SCRIMP**
Ferraro, G., Fournié, M. and Haine, G. (2024). *Simulation and control of interactions in multi-physics, a Python package for port-Hamiltonian systems*. IFAC-PapersOnLine.
- **Implicit port-Hamiltonian models**
Bendimerad-Hohl, A., Haine, G., Lefèvre, L. and Matignon, D. (2026). *Structure-preserving space discretization of differential and nonlocal constitutive relations for port-Hamiltonian systems*. Journal of Computational Physics.

- ① Introduction
 - Motivation
 - Bibliography

- ② Viscous sloshing in a fixed domain
 - Physics dynamics
 - Port-Hamiltonian modeling
 - Structure-preserving space discretization
 - Numerical results - SCRIMP

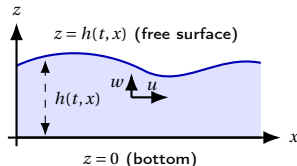
- ③ Viscous sloshing in a moving tank
 - From physics to PH modeling
 - Structure-preserving space discretization

- ④ Conclusion

Outline

- 1 Introduction
- 2 Viscous sloshing in a fixed domain
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Derivation of the viscous Saint-Venant model



- Negligible surface tension.

Parameters and variables

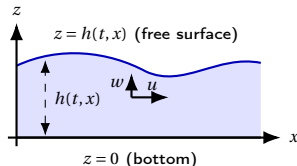
ρ fluid density, μ dynamic viscosity, b is the tank width, a is the tank length, g gravity $h(t,x)$ fluid depth, $u(t,x,z)$ horizontal velocity, $w(t,x,z)$ vertical velocity, $p(t,x,z)$ pressure, $\bar{u}(t,x)$ depth-averaged velocity.

$$\bar{u}(t,x) = \frac{1}{h(t,x)} \int_0^{h(t,x)} u(t,x,z) dz.$$

Main assumptions

- Incompressible Newtonian viscous fluid.
- Shallow-water regime: $\varepsilon = \frac{H}{a} \ll 1$.
- Hydrostatic pressure.
- Flat, impermeable and free-slip bottom.

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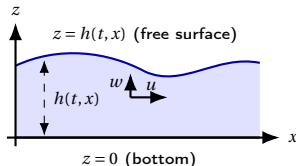
2-D incompressible Navier–Stokes

$$\begin{cases} \partial_x u + \partial_z w = 0, \\ \rho(\partial_t u + u\partial_x u + w\partial_z u) = -\partial_x p + \partial_x \sigma_{xx} + \partial_z \sigma_{xz}. \end{cases}$$

$$\sigma_{xx} = 2\mu\partial_x u, \quad \sigma_{xz} = \mu(\partial_z u + \partial_x w)$$

$\Downarrow$ depth integration
+ shallow-water approximation

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Viscous Saint–Venant model

Mass conservation

$$\partial_t h + \partial_x (h\bar{u}) = 0,$$

Horizontal momentum balance

$$\partial_t (h\bar{u}) + \partial_x \left(h\bar{u}^2 + \frac{g}{2} h^2 \right) = \nu \partial_x (h\partial_x \bar{u}).$$

$$\nu = \frac{\mu}{\rho} \quad (\text{kinematic viscosity})$$

Physics dynamics

Consider the 1D viscous shallow-water equations on the spatial domain $I = (-\frac{a}{2}, \frac{a}{2})$:

$$\begin{cases} \partial_t h = -\partial_x(hu) & t \geq 0, x \in]-\frac{a}{2}, \frac{a}{2}[\\ \partial_t(hu) = -\partial_x\left(hu^2 + \frac{1}{2}gh^2\right) + 2\nu h \partial_x(\partial_x u) & t \geq 0, x \in]-\frac{a}{2}, \frac{a}{2}[\end{cases}$$

Assumption. The fluid height remains strictly positive:

$$h(t, x) \geq h_{\min} > 0, \quad \forall (t, x) \in [0, T] \times I.$$

Using the continuity equation to eliminate $\partial_t h$ from the momentum equation, the latter can be equivalently rewritten as

$$\partial_t u = -\partial_x \left(\frac{u^2}{2} + gh \right) + \frac{2\nu}{h} \partial_x (h \partial_x u) \quad (1)$$

Hamiltonian (total energy)

$$H(h, u) = \int_I \underbrace{\frac{1}{2} \rho b h u^2}_{\text{kinetic energy}} + \underbrace{\frac{1}{2} \rho b g h^2}_{\text{potential energy}} dx$$

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Port-Hamiltonian formulation

Energy variables $\alpha_h = bh$ fluid section area, $\alpha_u = \rho u$ fluid momentum density.

The Hamiltonian in these variables reads

$$H(\alpha_h, \alpha_u) = \int_I \left(\frac{\alpha_h \alpha_u^2}{2\rho} + \frac{\rho g}{2b} \alpha_h^2 \right) dx \quad (2)$$

Co-energy variables $e_h = \frac{\alpha_u^2}{2\rho} + \frac{\rho g}{b} \alpha_h$ total pressure, $e_u^0 = \frac{\alpha_h \alpha_u}{\rho}$ volumetric flow.

The dynamics in these variables reads

$$\begin{pmatrix} \partial_t \alpha_h \\ \partial_t \alpha_u \end{pmatrix} = \begin{pmatrix} 0 & -\partial_x \cdot \\ -\partial_x \cdot & \frac{2}{\alpha_h} \partial_x \left(\rho \alpha_h \partial_x \frac{1}{\alpha_h} \cdot \right) \end{pmatrix} \begin{pmatrix} e_h \\ e_u^0 \end{pmatrix} \quad (3)$$

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Recall the assumption $\alpha_h(t, x) > 0$, $(t, x) \in [0, T] \times I$. We introduce the inner product $\langle f, g \rangle_{\alpha_h} := \int_I f \alpha_h g dx$, and denote by $L_{\alpha_h}^2(I)$ the corresponding Hilbert space. We consider $\alpha_h, e_h \in L^2(I)$, $\alpha_u, e_u \in L_{\alpha_h}^2(I)$.

Consequently $e_u = \delta_{\alpha_u}^{\alpha_h} H = \frac{\alpha_u}{\rho} = u \implies \partial_t \alpha_h = -\partial_x(\alpha_h e_u)$

Dynamics in co-energy variables e_h, e_u

$$(3) \implies \begin{pmatrix} \partial_t \alpha_h \\ \partial_t \alpha_u \end{pmatrix} = \begin{pmatrix} 0 & -\partial_x(\alpha_h \cdot) \\ -\partial_x \cdot & \frac{2\rho}{\alpha_h} \partial_x(\alpha_h \partial_x \cdot) \end{pmatrix} \begin{pmatrix} e_h \\ e_u \end{pmatrix} \quad (4)$$

Interconnection J & dissipative R operators

$$\underbrace{J(\alpha_h) = -J(\alpha_h)^*}_{\text{formally skew-adjoint}} = \begin{pmatrix} 0 & -\partial_x(\alpha_h \cdot) \\ -\alpha_h \partial_x & 0 \end{pmatrix}, \quad \underbrace{R(\alpha_h) = R(\alpha_h)^*}_{\text{formally Self-adjoint}} = \begin{pmatrix} 0 & 0 \\ 0 & -2\rho \partial_x(\alpha_h \partial_x \cdot) \end{pmatrix} \geq 0$$

This yields the state-space Port-Hamiltonian system

$$\begin{pmatrix} 1 & 0 \\ 0 & \alpha_h \end{pmatrix} \partial_t \alpha = (J(\alpha_h) - R(\alpha_h)) \delta_\alpha H(\alpha), \text{ where } \alpha = \begin{pmatrix} \alpha_h \\ \alpha_u \end{pmatrix}$$

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Dirac structure & resistive relation

$$\begin{pmatrix} \partial_t \alpha_h \\ \alpha_h \partial_t \alpha_u \\ f_r \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -\partial_x(\alpha_h \cdot) & 0 \\ -\alpha_h \partial_x & 0 & \partial_x \cdot \\ 0 & \partial_x & 0 \end{pmatrix}}_{J_D(\alpha_h) = -J_D(\alpha_h)^*} \begin{pmatrix} e_h \\ e_u \\ e_r \end{pmatrix}, \quad \underbrace{e_r = 2\rho \alpha_h f_r}_{\langle e_r, f_r \rangle \geq 0} \quad (5)$$

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Power balance equation

$$\underbrace{\frac{dH}{dt}(t)}_{\text{Stored power}} = - \underbrace{\left[(e_h - \rho \partial_x \alpha_u) \alpha_h e_u \right]_{-\frac{a}{2}}^{\frac{a}{2}}}_{\text{boundary power}} - \underbrace{2\rho \nabla \int_I \alpha_h (\partial_x e_u)^2 dx}_{\text{dissipated power} \geq 0} \quad (6)$$

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Partitioned finite element method

PFEM principle

- 1 Write the **weak formulation**;
- 2 Apply integration by parts to make the **boundary ports** appear;
- 3 Project the weak formulation on the **finite-dimensional** spaces.

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For all test functions φ_h , φ_u and φ_r

$$(5) \quad \begin{pmatrix} \partial_t \alpha_h \\ \alpha_h \partial_t \alpha_u \\ f_r \end{pmatrix} = \begin{pmatrix} 0 & -\partial_x(\alpha_h \cdot) & 0 \\ -\alpha_h \partial_x & 0 & \partial_x \\ 0 & \partial_x & 0 \end{pmatrix} \begin{pmatrix} e_h \\ e_u \\ e_r \end{pmatrix} \Rightarrow \begin{cases} \langle \partial_t \alpha_h, \varphi_h \rangle = -\langle \partial_x(\alpha_h e_u), \varphi_h \rangle, \\ \langle \alpha_h \partial_t \alpha_u, \varphi_u \rangle = -\langle \alpha_h \partial_x e_h, \varphi_u \rangle + \langle \partial_x e_r, \varphi_u \rangle, \\ \langle f_r, \varphi_r \rangle = \langle \partial_x e_u, \varphi_r \rangle. \end{cases}$$

Integration by parts on the 1st and 3rd equations:

$$\begin{cases} \langle \partial_t \alpha_h, \varphi_h \rangle = \langle \alpha_h e_u, \partial_x \varphi_h \rangle - \left[\alpha_h e_u \varphi_h \right]_{-\frac{a}{2}}^{\frac{a}{2}} \\ \langle f_r, \varphi_r \rangle = -\langle e_u, \partial_x \varphi_r \rangle + \left[e_u \varphi_r \right]_{-\frac{a}{2}}^{\frac{a}{2}}. \end{cases}$$

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Integration by parts on the 1st and 3rd equations:

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The energy, co-energy, and test functions of the same index are approximated using the same bases:

$$\begin{aligned} \alpha_h^d(t, x) &= \sum_{j=1}^{N_h} \alpha_h^j(t) \varphi_h^j(x) = \varphi_h(x) \cdot \underline{\alpha_h}(t), & e_h^d(t, x) &= \sum_{j=1}^{N_h} e_h^j(t) \varphi_h^j(x) = \varphi_h(x) \cdot \underline{e_h}(t) \text{ same for } \alpha_u \text{ and } e_u \\ f_r^d(t, x) &= \sum_{j=1}^{N_r} f_r^j(t) \varphi_r^j(x) = \varphi_r(x) \cdot \underline{f_r}(t), & e_r^d(t, x) &= \sum_{j=1}^{N_r} e_r^j(t) \varphi_r^j(x) = \varphi_r(x) \cdot \underline{e_r}(t) \end{aligned}$$

The semi-discrete system reads

$$\begin{pmatrix} M_h & 0 & 0 & 0 & 0 \\ 0 & M_u[\alpha_h^d] & 0 & 0 & 0 \\ 0 & 0 & M_r & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \dot{\underline{x}}_h \\ \dot{\underline{x}}_u \\ \underline{f}_r \\ -y_L \\ -y_R \end{pmatrix} = \begin{pmatrix} 0 & D[\alpha_h^d] & 0 & \alpha_h^- B_h^- & \alpha_h^+ B_h^+ \\ -D[\alpha_h^d]^\top & 0 & C & 0 & 0 \\ 0 & -C^\top & 0 & -B_r^- & -B_r^+ \\ -\alpha_h^- B_h^-{}^\top & 0 & B_r^-{}^\top & 0 & 0 \\ -\alpha_h^+ B_h^+{}^\top & 0 & B_r^+{}^\top & 0 & 0 \end{pmatrix} \begin{pmatrix} \underline{e}_h \\ \underline{e}_u \\ \underline{e}_r \\ u_L \\ u_R \end{pmatrix}.$$

where

$$(M_h)_{ij} = \int_I \varphi_h^i \varphi_h^j dx \quad (M_r)_{ij} = \int_I \varphi_r^i \varphi_r^j dx, \quad (M_u[\alpha_h^d])_{ij} = \int_I \alpha_h^d \varphi_u^i \varphi_u^j dx,$$

$$D[\alpha_h^d]_{ij} = \int_I \alpha_h^d \varphi_u^j \partial_x \varphi_h^i dx, \quad C_{ij} = \int_I \partial_x \varphi_r^j \varphi_u^i dx$$

$$(B_h^\pm)_i = \varphi_h^i(\pm \frac{a}{2}), \quad (B_r^\pm)_i = \varphi_r^i(\pm \frac{a}{2}), \quad (\alpha_h^\pm)_i = \alpha_h(\pm \frac{a}{2})$$

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$$(B_h^\pm)_i = \varphi_h^i(\pm \frac{a}{2}), \quad (B_r^\pm)_i = \varphi_r^i(\pm \frac{a}{2}), \quad (\alpha_h^\pm)_i = \alpha_h(\pm \frac{a}{2})$$

Discrete Hamiltonian

$$H^d(\underline{\alpha}_h, \underline{\alpha}_u) := H(\alpha_h^d, \alpha_u^d) = \frac{1}{2} \left(\frac{1}{\rho} \underline{\alpha}_u \cdot M_u[\alpha_h^d] \underline{\alpha}_u + \frac{\rho g}{b} \underline{\alpha}_h \cdot M_h \underline{\alpha}_h \right)$$

Constitutive relations

$$M_h \underline{e}_h = \frac{\rho g}{b} M_h \underline{\alpha}_h + N[\alpha_u^d] \underline{\alpha}_u, \quad M_u[\alpha_h^d] \underline{e}_u = \frac{1}{\rho} M_u[\alpha_h^d] \underline{\alpha}_u, \quad M_r \underline{e}_r = \rho \underline{v} M_r[\alpha_h^d] \underline{f}_r, \quad \text{where } (N[\alpha_u^d])_{i,j} = \frac{1}{2\rho} \int_I \alpha_u^d \varphi_u^j \varphi_u^i$$

Discrete power balance

$$\underbrace{\frac{dH^d}{dt}(t)}_{\text{Stored power}} = \underbrace{y_L u_L + y_R u_R}_{\text{boundary power}} - \underbrace{\underline{e}_r^\top M_r \underline{f}_r}_{\text{dissipated power} \geq 0} \quad (7)$$

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Numerical results - SCRIMP

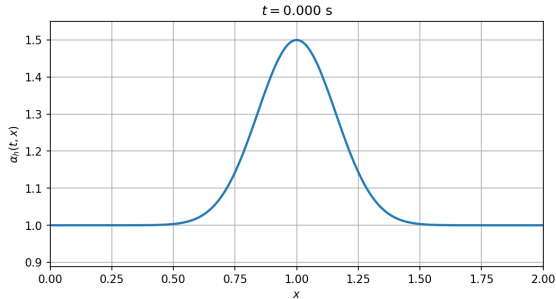
Physical parameters: $\rho = 10$, $\nu = 0.05$

Boundary conditions: no external control at the boundaries,

$$U_L(t) = U_R(t) = 0.$$

Initial conditions: the fluid is initially at rest, with a Gaussian free-surface deformation,

$$\alpha_u(0, x) = 0, \quad \alpha_h(0, x) = 2 + 0.5 e^{-20(x - \frac{a}{2})^2}.$$



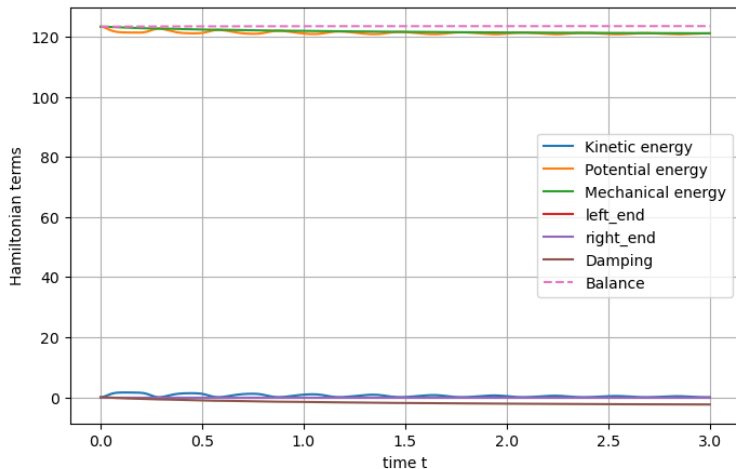
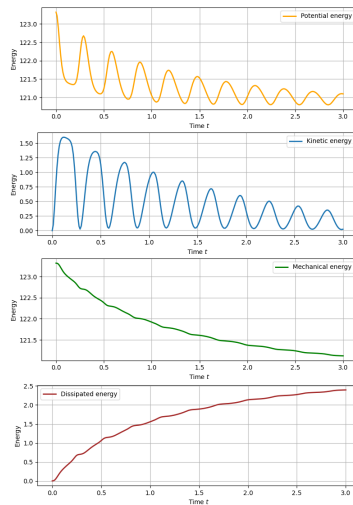


Figure: Evolution of Hamiltonian terms



$$\underbrace{\text{Potential energy} + \text{Kinetic energy}}_{H(t) : \text{Mechanical energy}} + \text{Dissipated energy} = \text{Constant}$$

Outline

- ① Introduction
- ② Viscous sloshing in a fixed domain
- ③ Viscous sloshing in a moving tank
 - From physics to PH modeling
 - Structure-preserving space discretization
- ④ Conclusion

Consider viscous fluid within a rigid tank under translation only:

$$\begin{cases} \partial_t h(t, x) &= -\partial_x(hu) & t \geq 0, \quad x \in]-\frac{a}{2}, \frac{a}{2}[, \\ \partial_t u(t, x) &= -\ddot{D} - \partial_x\left(\frac{u^2}{2} + gh\right) + \frac{1}{h}\partial_x(\nu h\partial_x u) & t \geq 0, \quad x \in]-\frac{a}{2}, \frac{a}{2}[, \\ m_T \ddot{D} &= F & t \geq 0, \end{cases} \quad (8)$$

where u denotes the fluid speed, measured relative to the tank, D the horizontal position of the tank, m_T the tank mass, and F is the sum of all forces applied to the tank.

The Hamiltonian is given by:

$$H(t) = \int_I \frac{1}{2} \left(\rho b h (u + \dot{D})^2 + \rho g b h^2 \right) dx + \frac{1}{2} m_T \dot{D}^2$$

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Energy variables

$$\alpha_h = bh \quad \text{fluid section area}$$

$$\alpha_2 = \rho(u + \dot{D}) \quad \text{fluid momentum density}$$

$$p = m_T \dot{D} + \int_I \alpha_h \alpha_2 dz \quad \text{total momentum}$$

$$D \quad \text{horizontal tank displacement}$$

Hamiltonian

$$H(\alpha_h, \alpha_2, p, D) = \int_I \frac{1}{2} \left(\frac{1}{\rho} \alpha_h \alpha_2^2 + \frac{\rho g}{b} \alpha_h^2 \right) dz + \frac{(p - \int_I \alpha_h \alpha_2 dz)^2}{2m_T}$$

Co-energy variables

$$e_1 = \frac{\alpha_2^2}{2\rho} + \frac{\rho g}{b} \alpha_h - e_p \alpha_2, \quad e_u = u, \quad e_p = \frac{1}{m_T} \left(p - \int_I \alpha_h \alpha_2 dx \right) = \dot{D}, \quad e_D = 0$$

PH reformulation

$$\begin{pmatrix} \partial_t \alpha_h \\ \partial_t \alpha_2 \\ \dot{p} \\ \dot{D} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -\partial_x \alpha_h \cdot & 0 & 0 \\ -\partial_x \cdot & 2\rho \alpha_h^{-1} \partial_x \alpha_h \partial_x \cdot & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{J-R} \begin{pmatrix} e_1 \\ e_u \\ e_p \\ e_D \end{pmatrix} + B F_{ext} \quad (9)$$

$$y_I = B^\top \begin{pmatrix} e_1 \\ e_u \\ e_p \\ e_D \end{pmatrix} = e_p = \dot{D}$$

where $B = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}^\top$, and $u_I = F_{ext}$ the input.

PH reformulation

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where $B = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}^\top$, and $u_I = F_{ext}$ the input.

Power balance equation

By considering the following boundary inputs and outputs

$$\begin{pmatrix} u_L \\ u_R \end{pmatrix} = \begin{pmatrix} \alpha_h e_u \left(-\frac{a}{2}\right) \\ -\alpha_h e_u \left(\frac{a}{2}\right) \end{pmatrix}, \quad \begin{pmatrix} y_L \\ y_R \end{pmatrix} = \begin{pmatrix} e_1 \left(-\frac{a}{2}\right) - \alpha_h \partial_x \alpha_2 \left(-\frac{a}{2}\right) \\ e_1 \left(\frac{a}{2}\right) - \alpha_h \partial_x \alpha_2 \left(\frac{a}{2}\right) \end{pmatrix}$$

Evolution of the Hamiltonian:

$$\frac{dH}{dt} = \underbrace{u_L y_L + u_R y_R}_{\text{boundary power}} + \underbrace{F_{ext} \dot{D}}_{\text{tank power}} - \underbrace{\langle e_r, f_r \rangle}_{\text{viscous dissipation}}$$

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Structure-preserving space discretization

Applying the same discretization procedure to the first two equations, the remaining ones being already finite-dimensional, yields:

$$\begin{pmatrix} M_h & 0 \\ 0 & M_u[\alpha_h^d] \end{pmatrix} \begin{pmatrix} \dot{\alpha}_h \\ \dot{\alpha}_2 \end{pmatrix} = \begin{pmatrix} 0 & D[\alpha_h^d] \\ -D[\alpha_h^d]^\top & 0 \end{pmatrix} - \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & R[\alpha_h^d] \end{pmatrix}}_{R^d = (R^d)^\top \geq 0} \begin{pmatrix} \underline{e}_1 \\ \underline{e}_u \end{pmatrix} + \text{Boundary conditions}$$

$J^d = -(J^d)^\top$ $R^d = (R^d)^\top \geq 0$

Discrete constitutive relations

$$\begin{pmatrix} M_h & 0 \\ 0 & M_u \end{pmatrix} \begin{pmatrix} \underline{e}_1 \\ \underline{e}_u \end{pmatrix} = \begin{pmatrix} \frac{\rho g}{b} M_h & -\frac{p - \alpha_h \cdot Q \alpha_2}{m_T} Q + A \\ -\frac{p - \alpha_h \cdot Q \alpha_2}{m_T} Q^\top + 2A^\top & 0 \end{pmatrix} \begin{pmatrix} \alpha_h \\ \alpha_2 \end{pmatrix}$$

Structure-preserving space discretization

Applying the same discretization procedure to the first two equations, the remaining ones being already finite-dimensional, yields:

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Discrete power balance

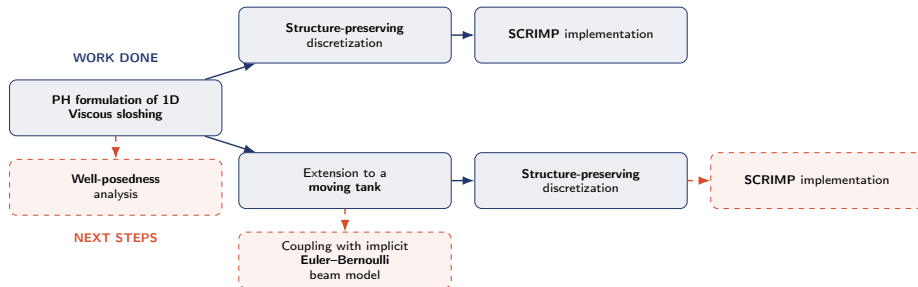
$$\frac{dH^d}{dt} = \underbrace{u_L y_L + u_R y_R}_{\text{boundary power}} + \underbrace{F_{\text{ext}} \nabla_p H^d}_{\text{tank power}} - \underbrace{\underline{e}_u^\top R[\alpha_h^d] \underline{e}_u}_{\text{viscous dissipation}} \quad (10)$$

⇒ The continuous power structure is preserved.

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Conclusion and perspectives

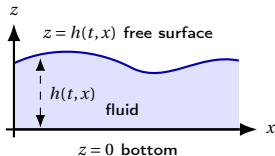


Upcoming activities

- **December 2026:** Research visit at ISAE-SUPAERO, Toulouse, focused on numerical simulations and a conference paper preparation.
- **January 2027:** Submission of a paper to LHMNLC conference (June 2027).
- **February 2027:** Port-Hamiltonian Symposium, Wuppertal: *Discrete Stokes-Dirac Structures and Finite Elements*.

THANK YOU!

Mass conservation



Assumptions

- Incompressible Newtonian viscous fluid.
- Shallow-water regime: $\varepsilon = \frac{H}{a} \ll 1$.
- Hydrostatic pressure.
- Depth-averaged horizontal velocity.
- Flat, impermeable and free-slip bottom.
- Negligible surface tension.

Mass conservation for an incompressible fluid,

$$\nabla \cdot \mathbf{v} = 0 \quad \Rightarrow \quad \partial_x u + \partial_z w = 0.$$

Integrating over the fluid depth $z \in [0, h(t, x)]$, using the condition $w(0) = 0$, and

$$w(h) = \partial_t h + u(h) \partial_x h,$$

gives

$$\partial_t h + \int_0^h \partial_x u \, dz + u(h) \partial_x h = 0.$$

By Leibniz's rule,

$$\int_0^h \partial_x u \, dz + u(h) \partial_x h = \partial_x \left(\int_0^h u \, dz \right),$$

hence

$$\partial_t h + \partial_x \left(\int_0^h u \, dz \right) = 0.$$

Introducing the depth-averaged velocity

$$\bar{u}(t, x) = \frac{1}{h(t, x)} \int_0^{h(t, x)} u(t, x, z) \, dz, \quad \int_0^h u \, dz = h \bar{u},$$

we obtain

Mass conservation

$$\partial_t h + \partial_x (h \bar{u}) = 0.$$

Horizontal momentum balance

Horizontal momentum equation

$$\rho(\partial_t u + u\partial_x u + w\partial_z u) = -\partial_x p + \partial_x \sigma_{xx} + \partial_z \sigma_{xz}.$$

Integrating over the fluid depth $z \in [0, h]$, and calculating term by term:

Hydrostatic pressure $p = \rho g(h - z)$

$$-\int_0^h \partial_x p \, dz = -\rho g h \partial_x h = -\rho \partial_x \left(\frac{gh^2}{2} \right).$$

Viscous stresses

$$\int_0^h \partial_x \sigma_{xx} \, dz = \partial_x \left(\int_0^h \sigma_{xx} \, dz \right) - \sigma_{xx}(h) \partial_x h,$$

$$\int_0^h \partial_z \sigma_{xz} \, dz = \sigma_{xz}(h) - \sigma_{xz}(0).$$

Therefore,

$$\rho \left[\partial_t \int_0^h u \, dz + \partial_x \int_0^h u^2 \, dz \right] = -\rho \partial_x \left(\frac{gh^2}{2} \right) + \partial_x \left(\int_0^h \sigma_{xx} \, dz \right) - \sigma_{xx}(h) \partial_x h + \sigma_{xz}(h) - \sigma_{xz}(0)$$

Boundary conditions

At the free surface: $-\sigma_{xx}(h) \partial_x h + \sigma_{xz}(h) = 0$, and at the bottom: $\sigma_{xz}(0) = \kappa u(0)$. For a frictionless bottom ($\kappa = 0$): $\sigma_{xz}(0) = 0$.

Thus,

$$\rho \left[\partial_t(hu) + \partial_x(hu^2) \right] = -\rho \partial_x \left(\frac{gh^2}{2} \right) + \partial_x \left(\int_0^h \sigma_{xx} \, dz \right).$$

Using

$$\sigma_{xx} = 2\mu \partial_x u, \quad u(t, x, z) \simeq u(t, x),$$

gives

$$\int_0^h \sigma_{xx} \, dz \simeq 2\mu h \partial_x u.$$

Viscous Saint-Venant momentum equation

$$\partial_t(hu) + \partial_x \left(hu^2 + \frac{gh^2}{2} \right) = 2\nu \partial_x(h \partial_x u) \quad \nu = \frac{\mu}{\rho}.$$

Proof of power balance

Two intergration by parts yield to the result

$$\begin{aligned}
 \frac{dH}{dt} &= \langle \partial_t \alpha_h, e_h \rangle_{L^2} + \langle \partial_t \alpha_u, e_u \rangle_{L^2_{\alpha_h}} \\
 &= - \left[e_h \alpha_h e_u \right]_{-\frac{a}{2}}^{\frac{a}{2}} - \int \underbrace{\mathbf{v} \rho \alpha_h \partial_x e_u}_{=\mathbf{v} \alpha_h \partial_x \alpha_u = e_r} \underbrace{\partial_x e_u}_{f_r} dx + \left[\rho \mathbf{v} \alpha_h e_u^h \partial_x e_u \right]_{-\frac{a}{2}}^{\frac{a}{2}} \\
 &= - \left[e_h \alpha_h e_u \right]_{-\frac{a}{2}}^{\frac{a}{2}} - \langle e_r, f_r \rangle_{L^2} + \left[\mathbf{v} \alpha_h e_u \partial_x \alpha_u \right]_{-\frac{a}{2}}^{\frac{a}{2}} \\
 &= - \left[(e_h - \mathbf{v} \partial_x \alpha_u) \alpha_h e_u \right]_{-\frac{a}{2}}^{\frac{a}{2}} - \langle e_r, f_r \rangle_{L^2}
 \end{aligned}$$

We can identify power and energy ports, e.g., as follows:

$$u = \begin{pmatrix} u_L \\ u_R \end{pmatrix} = \begin{pmatrix} e_u(-\frac{a}{2}) \\ -e_u(\frac{a}{2}) \end{pmatrix}, \quad y = \begin{pmatrix} y_L \\ y_R \end{pmatrix} = \begin{pmatrix} \alpha_h(e_h(-\frac{a}{2}) - \mathbf{v} \partial_x \alpha_u(-\frac{a}{2})) \\ \alpha_h(e_h(\frac{a}{2}) - \mathbf{v} \partial_x \alpha_u(\frac{a}{2})) \end{pmatrix}$$

Hence

$$\boxed{\frac{dH}{dt} = u \cdot y - \langle f_r, e_r \rangle}$$

Structure-preserving discretization

Infinite-dimensional pH system

PDE with distributed inputs:

$$\frac{\partial \alpha}{\partial t}(x, t) = (J - R) \frac{\delta H}{\delta \alpha} + \mathcal{B}u(x, t),$$

$$y(x, t) = \mathcal{B}^* \frac{\delta H}{\delta \alpha}.$$

where J is formally skew-adjoint, and R is formally self-adjoint and nonnegative.

Boundary conditions:

$$u_{\partial}, \quad y_{\partial}$$

Power balance:

$$\dot{H} = - \underbrace{\langle \delta_{\alpha} H, R \delta_{\alpha} H \rangle_I}_{\text{dissipated power}} + \underbrace{\int_{-\frac{a}{2}}^{\frac{a}{2}} u_{\partial} \cdot y_{\partial} + \int_I u \cdot y}_{\text{boundary+external power}}$$

Structure-preserving discretization

Resulting ODE:

$$\dot{\alpha}_d = (J_d - R_d) \nabla H^d + B_I u + B_{\partial} u_{\partial},$$

$$y = B_I^{\top} \nabla H^d,$$

$$y_{\partial} = B_{\partial}^{\top} \nabla H^d.$$

where J_d is skew-symmetric matrix, and R_d is positive semi-definite.

Discretized Hamiltonian:

$$H^d := H(\alpha = \alpha_d).$$

Power balance:

$$\dot{H}^d = - \underbrace{\nabla H^d{}^{\top} R_d \nabla H^d}_{\text{dissipated power}} + \underbrace{u_{\partial}^{\top} y_{\partial} + u^{\top} y}_{\text{boundary+external power}}$$

Frame Title

Choise of p

The choice of the variable p is made due to the classical transition [1] from Lagrangian to Hamiltonian mechanics. Indeed, regarding D as the position of the mechanical system (tank), its conjugate momentum is defined by

$$p = \frac{\partial L}{\partial \dot{D}},$$

where L denotes the Lagrangian of the coupled fluid-tank system. By a direct calculation, we obtain

$$p = m_T \dot{D} + \int_I \rho b h (u + \dot{D}) dz = m_T \dot{D} + \int_I \alpha_h \alpha_2 dz.$$

Thus, p is the generalized momentum conjugate to D and represents the total horizontal momentum of the coupled fluid-tank system.

Lemma

The operator defined on $\mathcal{D}(R_{22}) = \{f \in L^2_{\alpha_h} / \partial_x(\mathbf{v} \alpha_h \partial_x f) \in L^2\}$ by

$$\begin{aligned} R_{22} : \mathcal{D}(R_{22}) &\rightarrow L^2 \\ f &\mapsto -\partial_x(\mathbf{v} \rho \alpha_h \partial_x f) \end{aligned}$$

is formally self adjoint and nonnegative.

The previous results lead to the following modulated dirac structure

$$\begin{pmatrix} \partial_t \alpha_h \\ \partial_t \alpha_2 \\ \dot{p} \\ f_r \\ -\dot{D} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -\partial_x \alpha_{h\cdot} & 0 & 0 & 0 \\ -\partial_x \cdot & 0 & 0 & \alpha_h^{-1} \partial_x \cdot & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & \partial_x \cdot & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}}_J \begin{pmatrix} e_1 \\ e_u \\ e_p \\ e_r \\ F_{ext} \end{pmatrix} \quad (1.4)$$